Crude temperature calibration from the weekend run

Crystal **infi2439_SC_1** (SC-cut) · run `qcm_run_2026-10-30_074609` · Fri 30 Oct 08:46 → Mon 2 Nov 11:03 CET (74.3 h) · natural lab temperature only · analysis 2 Nov 2026

Summary

- **A crude calibration is possible, but only of the local slopes.** The building's weekend set-back moved the PT100 by just 0.96 °C (21.83–22.79 °C). The new profile **infi2439_SC_1_wkcal** is therefore linear and valid for about 21.5–23 °C.
- **The thermometer is the lower of two modes, by choice.** The temperature PLL is locked to a resonance at 6566.2 kHz, 3.6 kHz below the main B-mode, because its amplitude grows with temperature (+3–4 %/°C) while the main mode's collapses (7× weaker at 45 °C than at 9 °C). It responds at −91.2 Hz/°C, about 0.6× the main mode. The old coefficients were made for a different crystal and mode, so the displayed temperature showed only 0.52× the real swing; the new calibration tracks the PT100 to 4 mK rms once the 13 min thermal lag is allowed for.
- **Mass-mode tempco: 5.0 ± 1.4 Hz/°C** (+0.83 ppm/°C). That agrees with the Oct 28 cooling run (+5.0) and the old house curve (+4.9). Fitting fM directly against the room thermometer gives 12.5 Hz/°C, 2.5× too large, because the drift and the electronics heating also follow the room.
- **Weekend error band** (linear drift removed): 5.85 Hz uncompensated → 3.90 Hz with the old coefficients → **2.33 Hz** (31 ng/cm², 16 pm Au) with the new calibration, and 1.49 Hz if the board temperature is also corrected. Most of what remains sits in the two warming periods and at the load events.
- **Not compensated**, largest first: a steady drift of −2.3 Hz/day (≈ 30 ng/cm² per day of apparent mass uptake); electronics (fM follows the board temperature at ≈ +0.9…2.0 Hz/°C, including a 1.1 Hz step at Fri 17:25 and short 0.3–0.5 Hz spikes at load changes on the Pitaya); a rise of up to 1 Hz on Monday morning that is not a slope error (the two warm-ups alone imply 2.4 and 7.8 Hz/°C) and is probably humidity; and the calibration's own slope uncertainty (±1.4 Hz per °C of excursion). Noise is negligible (2 mHz).
- **Caution:** the profile is valid only while the temperature PLL stays on the 6566 kHz mode, so its lock start (freq_temp) is set to 6 566 157 Hz instead of 6 570 000 Hz. Depositions heat the crystal 5–10 °C, so for them do a Peltier calibration over ≥ 20 °C on this same 6566 kHz mode.

1 The data

The run log is the merged 14-part CSV downloaded from the Pitaya. Everything below uses 10 s medians (26 745 bins); the median removes a few single-sample glitches. The Allan deviation uses every sample. The run's own compensated column was recomputed with the instrument's `TempCompAlgorithm` and the old coefficients, and matches the logged values to 1.2 mHz. Every compensated number in this report therefore comes from the same code the instrument runs.

Item	Value
Samples	2 397 109 at 9.05 Hz, no gaps > 2 s; both PLLs locked 100 % of the time
Thermometer	PT100 on the LTC2986, no faults all run: 21.83–22.79 °C (span 0.96 °C), 1 s updates, ~0.01 °C noise
Board temperature	TSic 501F cold-junction sensor on the V2 board: 30.49–32.30 °C. The thermocouple input faulted all run (nothing connected).
Mass mode fM	5.98185 MHz, span 11.6 Hz (drift + temperature)
Temperature mode fT	6.56616 MHz, span 87 Hz
Coefficients during the run	Old crystal's house curve (the infi2439_SC_1 profile still carries the defaults); not changed mid-run
Ramp rates	natural only: at most +0.13 / −0.09 °C per hour

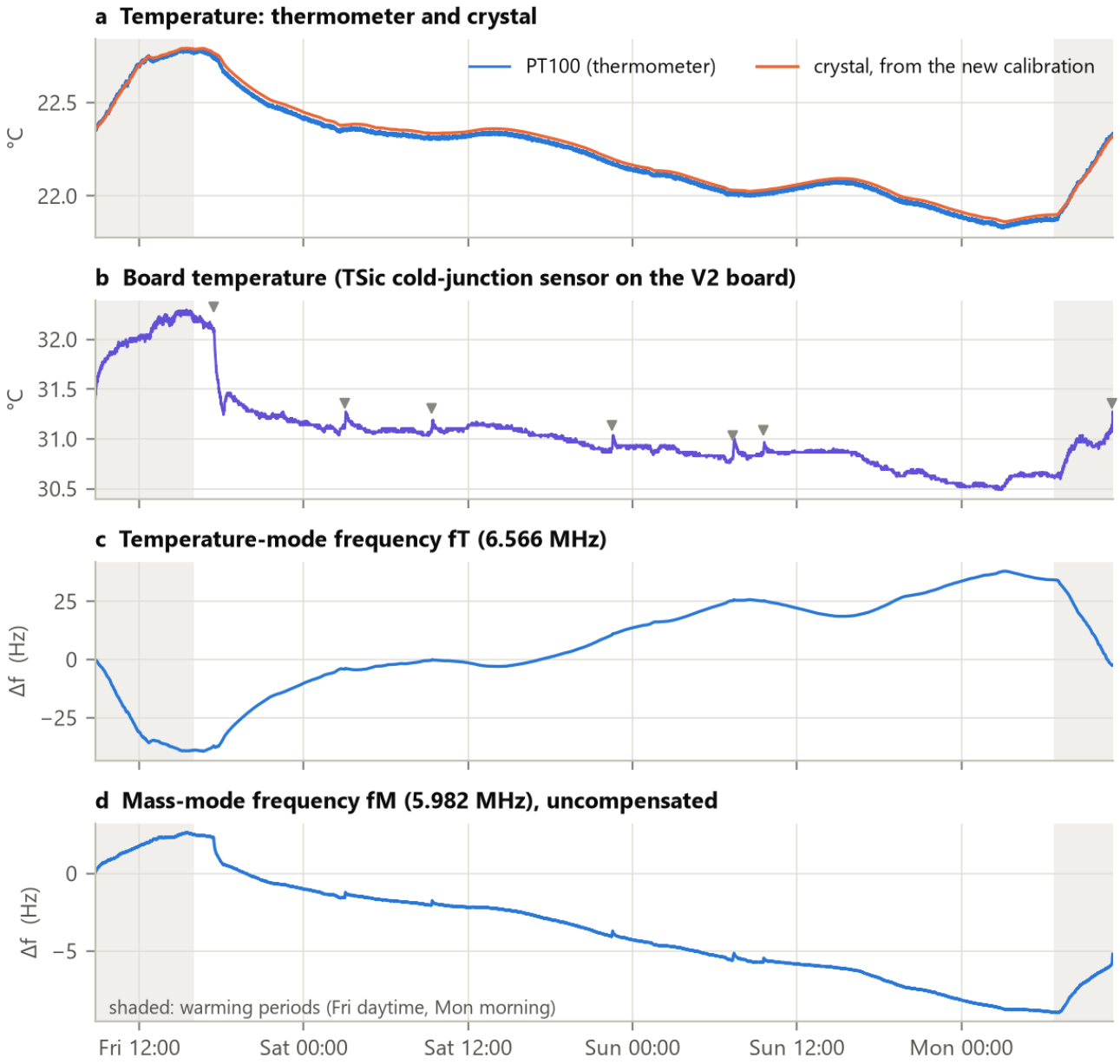


Figure 1. The weekend. (a) PT100, and the crystal temperature from the new calibration (it lags the PT100 by about 13 min). (b) Board temperature; arrows mark abrupt jumps, the large one on Fri 17:25. (c) fT falls when the crystal warms (negative tempco). (d) fM: a slow downward drift with the temperature response on top. Shaded: the two warming periods, Friday daytime and Monday morning.

2 The thermometer mode

Measured against the PT100, the weekend fT moves −91.2 Hz/°C (−13.9 ppm/°C). The calibration points taken on this crystal on Oct 28 (points 1–12, Peltier, PT100 source) give −157 Hz/°C between 14 and 25 °C, close to the textbook SC-cut B-mode (−25 ppm/°C = −165 Hz/°C). The B-mode region contains two resonances 3.6 kHz apart (Figure 2). Points 1–12 were taken on the strong upper one (fT ≈ 6569–6571 kHz). After the sweeps at 16:37–16:43 UTC the lock was deliberately moved to the lower one, and point 13 (16:54 UTC, 6566.1 kHz) and the whole weekend (6566.12–6566.20 kHz) are on it.

The reason is amplitude. The main mode fades as the crystal heats: during the Oct 28 cooling run its locked amplitude went from 157 at 45 °C to 1112 at 9 °C, a factor of 7, the same collapse seen on earlier crystals. It would threaten the lock, and the auto lock-detect threshold, at exactly the temperatures a deposition reaches. The lower mode does the opposite: over the weekend its amplitude rose 3.4–4.3 % per °C (correlation 0.97 with the PT100). The price is sensitivity, 0.6× the main mode, which costs nothing measurable here: fT's noise floor of 3 mHz is 0.03 mK on this mode.

The two peaks also respond differently to temperature in the sweeps: between 16:37 and 16:43, while the crystal warmed back towards room temperature, the lower peak moved about 0.7× as far as the main one (100 Hz resolution),

consistent with the 0.6 ratio of the slopes. So the smaller slope belongs to the mode; the thermometer is fine. Two consequences follow:

- The new calibration belongs to the 6566 kHz mode. If the PLL relocks on the main mode (6569.6 kHz at 22 °C), the temperature scale is wrong by a factor of about 1.7, and so is the compensation.
- The old profile's lock start of 6 570 000 Hz points at the main mode. The new profile therefore sets freq_temp to the weekend value, 6 566 157 Hz. At ±10 °C the lower mode stays within about 1 kHz of that start, while the main mode is 3.6 kHz away.

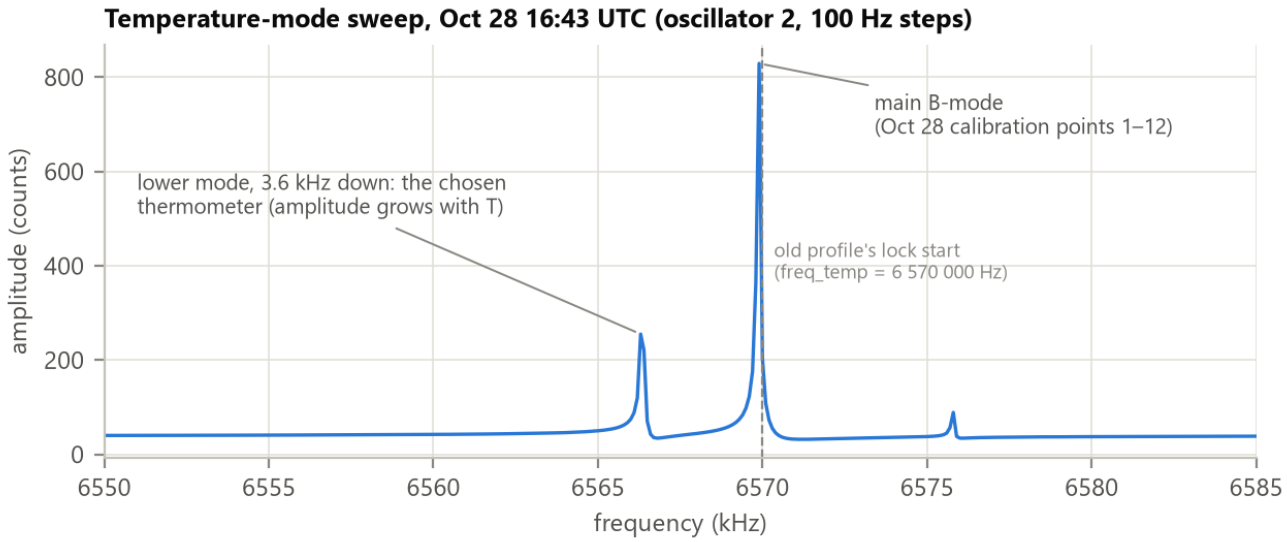


Figure 2. Temperature-mode sweep on Oct 28 at room temperature, 11 min before calibration point 13. The weekend lock (6566.12–6566.20 kHz) is on the left peak: smaller here, but unlike the main peak it grows as the crystal heats. A third, very weak resonance sits at 6575.8 kHz.

3 Temperature scale: thermometer → crystal

fT was fitted as a linear function of the PT100 passed through a first-order lag, plus a board-temperature term and a linear drift. Scanning the lag gives a clear optimum at **τ = 13 min**: the crystal follows the PT100 with 13 min delay (Figure 3b). That fit gives **dfT/dT = −91.2 ± 1.9 Hz/°C** with 0.38 Hz rms (4 mK). Without the lag the slope comes out 6 % lower and the residual is 3× larger. The uncertainty is a 6 h moving-block bootstrap, since the residuals are strongly autocorrelated.

The old coefficients assume about −181 Hz/°C, so the instrument displayed only 0.52× the real temperature swing, with errors up to 0.46 °C. With the new profile the displayed temperature follows the lagged PT100 with gain 0.995 and 4 mK rms (29 mK peak-to-peak; this is in-sample).

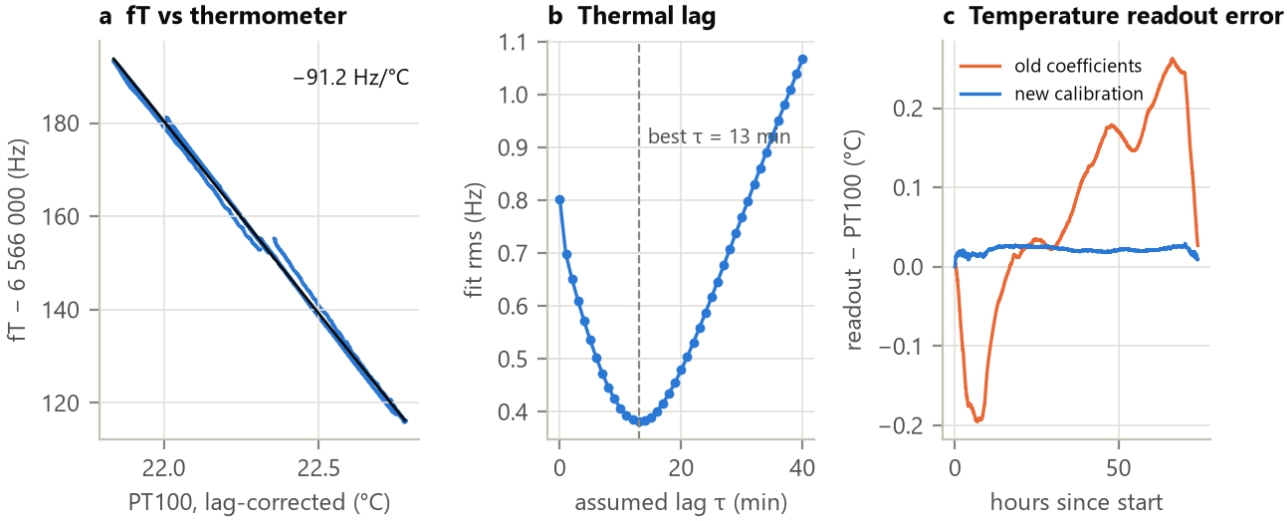


Figure 3. (a) fT against the lag-corrected PT100. (b) Fit residual against the assumed lag. (c) Displayed temperature minus the lagged PT100, for the old coefficients (as logged) and the new calibration.

4 The mass-mode temperature coefficient

This is the hard part: over the weekend, three things moved fM together, and all three follow the room.

- **Temperature of the crystal**: the effect to calibrate.
- **A slow drift** of about −2 Hz/day. Both modes drift together in close to the ratio that mass loading produces (fT/fM drift ratio 0.88; mass loading gives 1.10). The room cooled through most of the weekend, so this drift looks like temperature.
- **The electronics.** At Fri 17:25 both PLL amplitudes stepped by +0.3 % at once. The board temperature then fell 0.57 °C with a ~15 min time constant, and fM followed the board, not the crystal: −1.2 Hz while the PT100 changed by 0.03 °C (Figure 4). Five later spikes show the same signature with the opposite sign, and so does the moment the run was stopped. This looks like a change in electrical load on the Pitaya (a client disconnecting, scheduled OS jobs) feeding through the hot linear 5 V regulator on the V2 board. The board warms with the room too, so in a naive fit its effect also lands in the temperature coefficient.

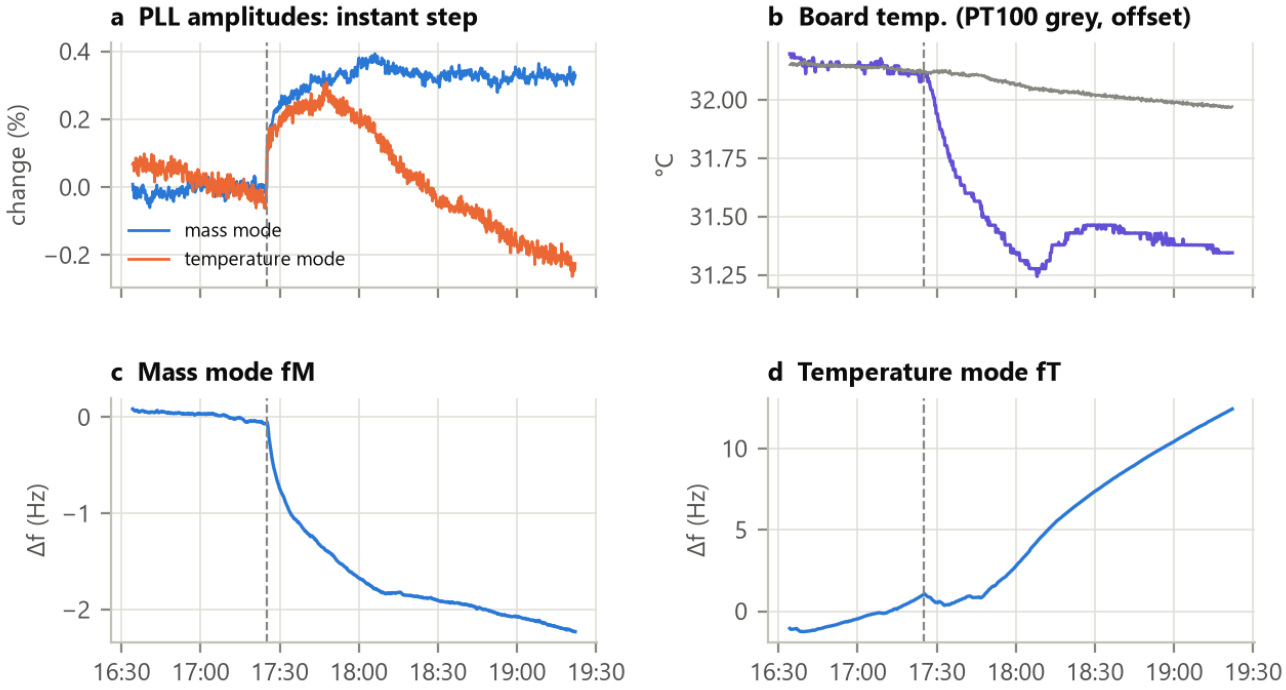


Figure 4. The Friday 17:25 event. The amplitudes step instantly; the board cools over ~40 min; fM follows the board temperature (≈ +2 Hz/°C); fT barely reacts, and the crystal temperature (PT100, grey in b) keeps its slow trend.

fM was therefore fitted against fT (the crystal's own thermometer, which needs no lag correction) together with a linear drift and the board temperature. Figure 5 shows how the answer depends on what the model includes and on which part of the weekend is used.

Model (fM fitted against)	fM/fT ratio	dfM/dT (Hz/°C)	board term (Hz/°C)	rms (Hz)
PT100 only (naive)	−0.151	12.5	—	0.76
fT + linear drift	−0.072 ± 0.013	6.5	—	0.30
fT + drift + board temperature (adopted)	**−0.0547 ± 0.015**	**4.99 ± 1.4**	0.95 ± 0.73	0.24
… with curved (quadratic) drift	−0.055 ± 0.010	5.1	1.52 ± 0.27	0.18
… 12 h high-pass instead of a drift term	−0.053	4.8	1.63	0.12

Once the drift and the board temperature are in the model, the three formulations agree to within 0.3 Hz/°C, at **5.0 Hz/°C**. The two independent references agree too: the Oct 28 cooling run (45 → 9 °C in 10 min, main B-mode; too fast for slow drifts to matter) gives +5.0 Hz/°C on average, and the old crystal's wide-range curve gives +4.9 Hz/°C at 22 °C. The Oct 28 figure averages over 9–45 °C; scaled to 22 °C with the house curve's shape it becomes ≈ +5.8 Hz/°C, within 1σ of the adopted value. The weekend cannot choose between them: with the board term included, the error band is 1.49 / 1.44 / 1.49 Hz for 5.0 / 5.8 / 6.5 Hz/°C.

The weekend data alone pin this number down only loosely. The estimate relies mainly on the two warming periods, and the quiet weekend on its own (10–70 h, a slow monotonic cool-down that is almost collinear with the drift) gives 1.6–3.6 Hz/°C. The two halves give 3.7 and 5.7. The bootstrap ±1.4 Hz/°C is therefore a fair statement of the uncertainty, and only the independent references justify the central value. With a 1 °C span nothing better is possible: slope uncertainty scales as 1/span, so a 20 °C Peltier calibration would shrink it about 20×.

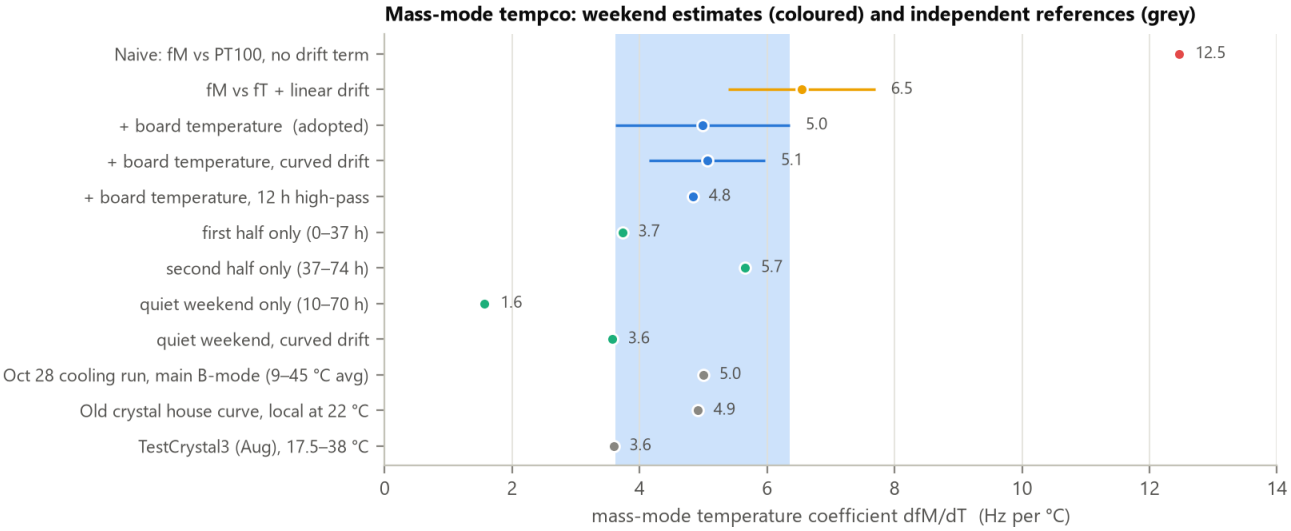


Figure 5. Every estimate of dfM/dT. Coloured: weekend fits (blue: adopted family, with bootstrap 1σ; green: sub-windows). Grey: independent references. Shaded: the adopted value ± 1σ.

5 The crude calibration: profile infi2439_SC_1_wkcal

The profile is linear (the quadratic and cubic terms are zero) because 1 °C of data cannot resolve curvature. Its temperature axis is the PT100's. Only the slope terms fM_1 and fT_1 enter the compensation. The constant terms just anchor the curves at the last hour of the run (T = 22.26 °C) for the GUI's crystal graphs. Lock starts, sensor constants and usage counters are carried over from infi2439_SC_1, except freq_temp (Section 2).

Field	Value	Meaning
fT_1	−91.1624 Hz/°C	temperature-mode slope (6566 kHz mode), ±1.9
fM_1	4.9859 Hz/°C	mass-mode slope = ratio × fT_1, ±1.4
fT_0 / fM_0	6568186.014 / 5981741.999 Hz	intercepts at 0 °C (display only)
fT_2, fT_3, fM_2, fM_3	0	not resolvable from a 1 °C span
freq_temp / freq_mass	6 566 157 / 5 981 853 Hz	lock starts: end-of-run frequencies; freq_temp is on the lower (6566 kHz) mode
Valid range	≈ 21.5–23 °C (crystal)	measured 21.85–22.8 °C

The file is `data/crystals/infi2439_SC_1_wkcal.json` in the repo. Loaded through the instrument's `CrystalManager` and run through `TempCompAlgorithm`, it reproduces this report's compensated series exactly. Set the run's ambient temperature to the PT100 reading at start so that the displayed temperature is absolute.

Outside the range the error grows by about ±1.4 Hz for every °C away from 22.3 °C (slope uncertainty). The missing curvature adds roughly 0.1 Hz × ΔT^2 more: an SC-cut mass mode bends by about −0.17 Hz/°C², and the lower mode's curvature is unknown. For a deposition that heats the crystal by 9 °C, expect up to about ±20 Hz (±270 ng/cm²), compared with ≈ 45 Hz uncompensated. That is useful, but far from the ~1 Hz this rig achieves with a proper calibration.

6 How well it compensates the weekend

Drift is apparent mass. A temperature compensation cannot remove it and should not, so each output is judged after removing its own best-fit linear drift. The remaining peak-to-peak deviation is the temperature error band (M1). The other metrics follow the scheme used since August: M2 = uncompensated ÷ compensated, M4 = M1 ÷ dfM/dT, M5 = drift. M3 (heating/cooling hysteresis) cannot be measured here: the only warming happened at the two ends of the run, during working hours, so it is confounded with everything in Section 7.

Output	M1 band	M1 in other units	M2	M4	M5 drift
Uncompensated fM	5.85 Hz	78 ng/cm² · 40 pm Au · 0.98 ppm	—	—	−3.57 Hz/day
Old coefficients (logged)	3.90 Hz	52 ng/cm² · 27 pm Au · 0.65 ppm	1.5×	0.78 °C	−2.91 Hz/day
New calibration	2.33 Hz	31 ng/cm² · 16 pm Au · 0.39 ppm	2.5×	0.47 °C	−2.27 Hz/day
New + board-temperature term	1.49 Hz	20 ng/cm² · 10 pm Au	3.9×	0.30 °C	—

The M1 band is over crystal temperatures 21.85–22.8 °C, across the whole weekend. Units: 13.3 ng/(cm²·Hz) (Inficon SC-cut) and 144.5 Hz/nm for gold.

The new calibration takes out the 4.6 Hz thermal component and cuts the band 2.5× (old coefficients: 1.5×). M2 is modest only because the weekend's thermal signal was small, and most of what remains is not crystal temperature (Section 7). Only the whole weekend can test the compensation. Between Fri 18:46 and Mon 06:46 the room cooled almost monotonically, so a drift line also takes out the temperature signal: there, uncompensated, old and new all show about 1 Hz (1.08 / 0.82 / 1.09 Hz). The warm-ups at the ends break that collinearity. The calibration also holds out of sample: fitted on one half and applied to the other, the residual is 0.25 / 0.28 Hz rms, against 0.24 Hz in-sample. The ratio fitted without the board term does slightly better in-sample (1.88 Hz), but it absorbs electronics drift into the crystal coefficient. It would over-correct whenever the crystal heats without the electronics (a deposition), so it was not adopted.

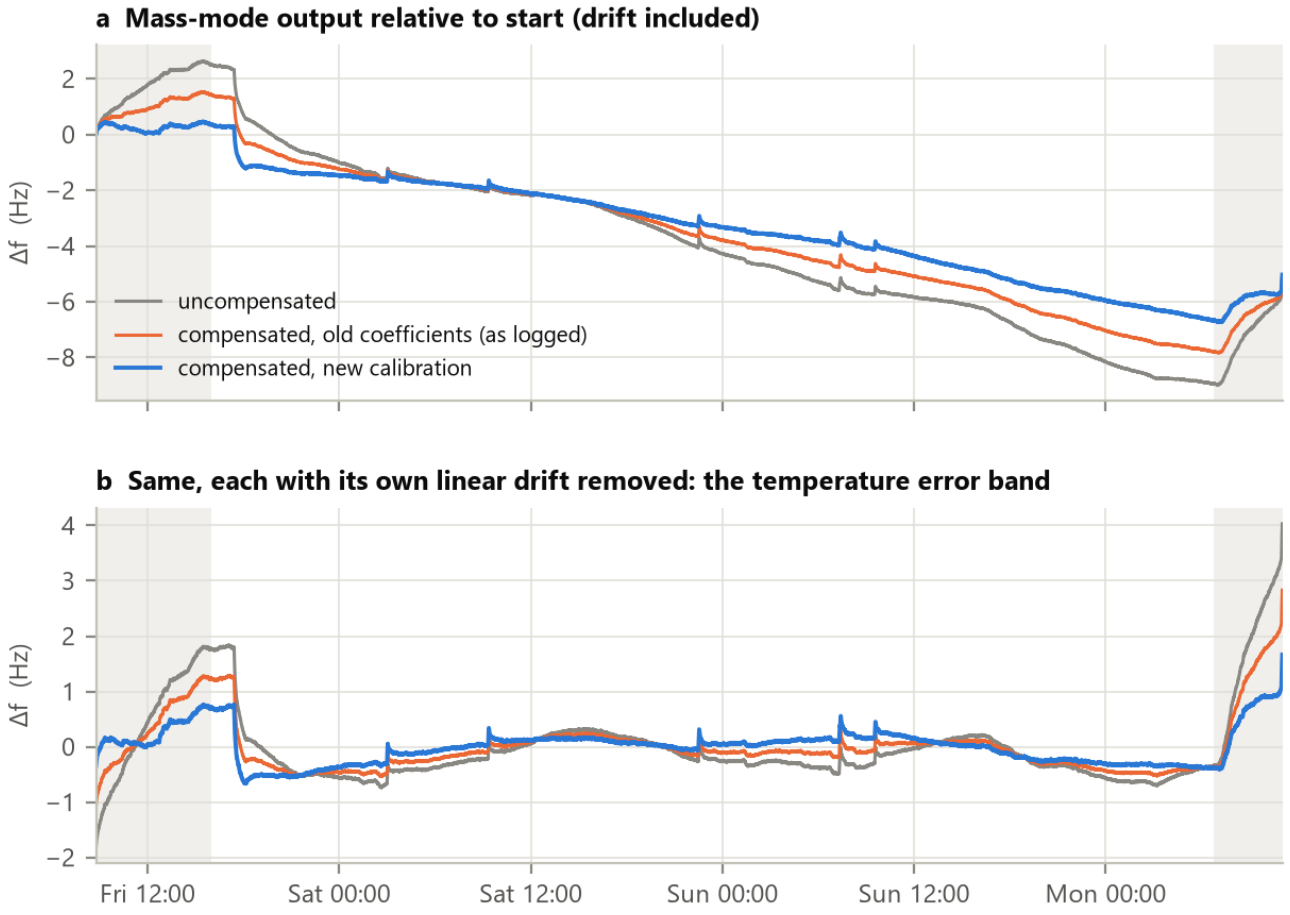


Figure 6. (a) Outputs relative to the start, drift included. (b) Each with its own linear drift removed. What remains of the new calibration's band sits mostly in the two warming periods (shaded) and at the load events.

7 What is not compensated

Effect	Size on the compensated output	Evidence	What would fix it

Drift (apparent mass uptake)	−2.27 Hz/day ≈ 30 ng/cm²/day; slightly accelerating (−1.6 → −2.2 Hz/day)	Both modes drift together (ratio 0.88; mass loading gives 1.10), so it is mass-like and the temperature readout stays flat. 5× the settled TestCrystal3 rate (−0.42 Hz/day).	Wait and re-measure; check for contamination; a dry or vacuum environment.
Electronics / board temperature	≈ +0.9…2.0 Hz per board °C; board swung 1.8 °C. Removing the term: 0.32 → 0.23 Hz rms	Fri 17:25 step: amplitudes step instantly, fM follows the board with ~15 min lag; five load spikes of 0.3–0.5 Hz.	Keep the hot linear regulator away from the front end (DC-DC with proper filtering); or correct fM in software with the TSic reading.
Warming periods (Friday, Mon morning)	up to +1 Hz on Monday morning, less on Friday, beyond the board term; 1.5 Hz p-p in those windows	Not a slope error: fitted alone, Friday's warm-up implies 2.4 Hz/°C and Monday's 7.8. Something environmental; adsorbed water following RH (not measured) is the prime suspect.	Log RH, or measure in vacuum or dry N2.
Calibration slope uncertainty	±1.4 Hz per °C of excursion: ±1.4 Hz across this weekend, ±12 Hz at 9 °C	Bootstrap and sub-window spread (Figure 5).	Peltier calibration over ≥ 20 °C.
Curvature / range	unknown; ≈ 0.1 Hz × ΔT^2 for an SC-cut mass mode	Linear profile, 1 °C of data.	Same.
Mode hop on the temperature PLL	would put the scale out by ×1.7	Two resonances 3.6 kHz apart (Figure 2).	Lock start on the calibrated mode; check fT after each lock.
Noise	2 mHz Allan minimum at ~6 s	Figure 8.	— (negligible)

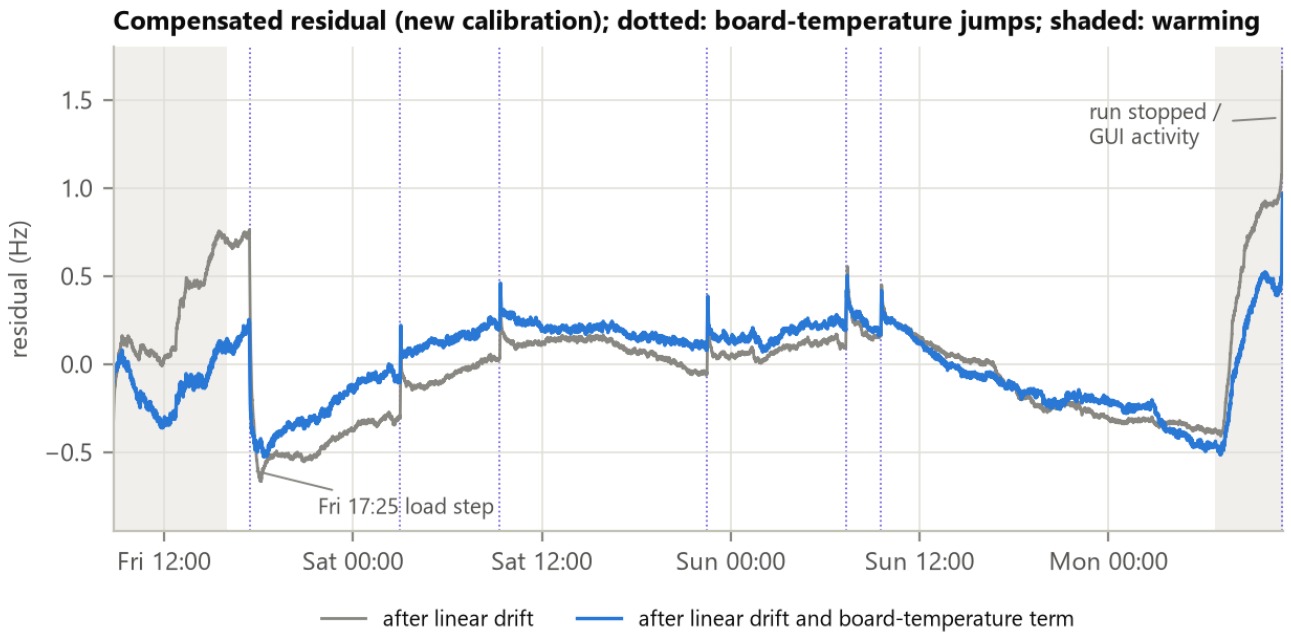


Figure 7. Residual of the new calibration after its linear drift (grey) and after drift plus board term (blue). Between the warming periods the blue curve stays within about ±0.5 Hz (0.22 Hz rms), and the jumps line up with board-temperature events. The final spike is when the run was stopped.

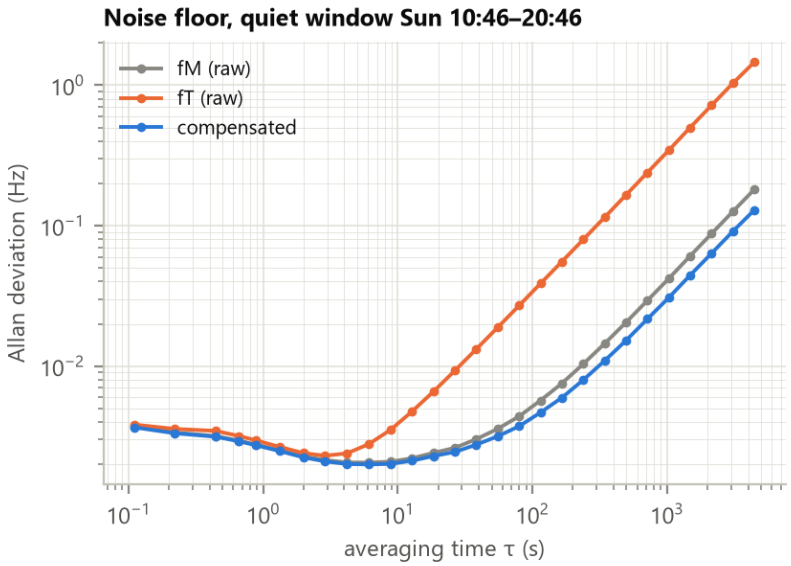


Figure 8. Allan deviation on a quiet 10 h window. The compensated output has the mass mode's noise, ~2 mHz (0.03 ng/cm²) at 1–10 s. Beyond a minute everything is slow systematics. fT rises faster because it carries the room's temperature wander, which is the signal it exists to measure.

8 Recommendations

- **Use infi2439_SC_1_wkcal for room-temperature work only** (≈ 21.5–23 °C), and check after every lock that fT is near 6 566 kHz (−91 Hz/°C) and not near 6 570 kHz.
- **Keep the lower mode as the thermometer.** Its amplitude grows with temperature while the main B-mode's collapses, and its 0.6× sensitivity costs nothing measurable. Pin it with the lock start (as the new profile does), and make sure every calibration point is taken on it.
- **Do a proper Peltier calibration on the 6566 kHz mode before a deposition**, following the method in use since August: ≥ 20 °C span centred on the operating range, 25–30 min soaks (the 13 min lag found here confirms that soak length), points in both directions. The CALIBRATE tab's settle detector currently accepts fT drift up to 10 Hz/min after a 60 s hold, looser than the 2 Hz/min rule; tighten it first.
- **Attack the electronics term at source.** The linear 5 V regulator on the V2 board (~14 W at full load) most likely turns load changes into board-temperature steps. Until it is replaced, avoid connecting or disconnecting clients during sensitive measurements. A board-temperature correction from the TSic is possible, but its coefficient is only known to ±50 %.
- **Add a humidity reading** (or measure in vacuum) before reading anything into the remaining ±0.5–1 Hz structure.
- **Re-check the drift** after another quiet day. At −2.3 Hz/day it dominates every long measurement and is 5× the previous crystal's.

Appendix A Method details

- **Binning.** 10 s medians of every channel. Single-sample glitches (an fT spike of 10.9 Hz at 65.46 h, an fM spike of 0.38 Hz at 20.19 h) are removed by the median.
- **Thermal lag.** PT100 → crystal modelled as a first-order low-pass $y' = (x - y)/\tau$, applied causally; τ scanned over 0–40 min in 1 min steps against the fit $fT = a + \beta \cdot lag(PT100) + e \cdot T_board + d \cdot t$. Best τ = 13 min.
- **Mass-mode ratio.** Ordinary least squares $fM = a + r \cdot fT + e \cdot T_board + d \cdot t$ on all bins. fT is the crystal's own thermometer, so no lag correction is needed. Uncertainties: 300 resamples of a moving-block bootstrap with 6 h blocks. Held-out check: fit r and e on one half; on the other half refit only the offset and drift.
- **Profile.** $fT_1 = \beta = -91.1624$ Hz/°C; $fM_1 = r \cdot \beta = 4.9859$ Hz/°C. The instrument's mass-free combination then yields $compensated = (fM - r \cdot fT)/(1 - \rho \cdot r)$ with $\rho = fT/fM \approx 1.098$, so only the ratio matters for mass, and β sets the temperature display.
- **Evaluation.** Every compensated series comes from the repo's TempCompAlgorithm.FreqToTemp, sample by sample, with the start values of the run. Drift removal: each series' own least-squares line over the window considered.
- **Event times** are CET (UTC+1). Load events (board temperature changing by more than 0.08 °C in 3 min): Fri 17:25 (down), Sat 03:00, Sat 09:20, Sat 22:30, Sun 07:19, Sun 09:33, Mon 11:00 (run stop).
- **Oct 28 references.** Calibration points from data/calibration/points.json on the Pitaya; sweeps qcm_sweep_2026-10-28_16*.csv; cooling run qcm_run_2026-10-28_162136.csv (PT100 45.4 → 8.9 °C, fM −183 Hz, fT +5193 Hz on the main B-mode).

Appendix B Profile file


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  "freq_temp": 6566157,
  "freq_virgin": 6000000.0,
  "fM_0": 5981741.998957444,
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